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Characteristics analysis of droplet transfer in laser-MAG hybrid welding process

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Highlights

- The motion characteristics of droplets in hybrid arc space are described.
- The action of metal vapor ejected from laser keyhole on droplet is analyzed.

- The laser plasma compresses the arc and leads to the increase of the electromagnetic resistance of the droplet transfer.

Abstract

The droplet transfer, plasma morphology and droplet morphology of laser arc hybrid welding process are collected and analyzed using high speed camera. The force magnitude and acceleration of motion droplets in the arc space are calculated by the image processing and mathematical calculations. The value and distribution of the recoil force from the metal vapor on droplets are calculated. The results show that the globular transfer mode corresponds to the welding current is approximate 180 A; the streaming transfer mode corresponds to the welding current is approximate 200 A. The laser has a compress effect on the arc, and the compression on the surface of the weld pool is stronger. The acceleration of the droplet detachment from wire is 70 m/s^2 and 50 m/s^2 for arc welding and laser arc hybrid welding separately. In the actual welding process, the reaction force of metal vapor on molten droplets is very small, when the distance between droplet and keyhole in the surface of the weld pool is 3 mm. The pressure difference on the surface between upper and lower of the droplet is great, which results the droplet coalescence and the transfer frequency slows down.

Previous

Next

Keywords

Laser-arc hybrid welding; High nitrogen steel; Metal droplet; Mechanics analysis

1. Introduction

The laser-arc hybrid welding process combining of a laser and an electrical arc sources has attracted considerable attentions recently due to its unique advantages compared with a single heat source welding [1]. The speed of hybrid welding is quite high and this process can result in deep penetration and good gap bridging. Therefore, the laser-metal inert gas welding/metal active gas welding (MIG/MAG) is finding applications in diverse industries like automotive, aerospace, shipbuilding, oil and pressure vessel industries, etc.

In recent years, many researchers have studied the coupling mechanism of different plasmas in laser-arc hybrid welding [2], [3], [4], and the influence of welding parameters on process stability, weld profile as well as weld penetration. Kutsuna and Chen [5] performed laser-MAG hybrid welding experiments to investigate the effects of process parameters on penetration depth and the interaction between laser plasma and MAG arc plasma. The results indicated that the penetration depth of hybrid welding is higher than those of laser or MAG welding only due to the laser plasma deflected towards the arc plasma which reduces the amount of laser energy loss. Wang et al. [6] captured the plasma images from different welding processes (Laser only, MIG only, and Laser-MIG hybrid) using a high speed camera. The experimental results showed that a curved channel was formed between the wire and laser keyhole during the laser-MIG hybrid welding process. The spectral intensities of the metal and shielding gas elements for the MIG-only welding and laser-MIG hybrid welding processes are significantly different. Liu and Chen [7] examined the shape, electron temperature and electron density of hybrid plasma during laser-arc hybrid welding. It was found that the distance between tungsten electrode and laser would influence the interaction between the laser and the arc. In addition, the electron temperature increases and electron density decreases when the keyhole is directly under the arc plasma due to the expansion caused by keyhole vapor colliding with the arc plasma. Gu et al. [8] investigated the physical interaction between the laser and arcs during laser-twin-arc hybrid welding process and analyzed the influence of the laser/arc heat source on the welding process. The experimental results indicated that the laser builds a conductive channel for the arcs that would affect the arc shape and stabilize the arc. The electron temperature will be significantly influenced by laser parameters.

The stability of hybrid welding process depends on droplet transfer mode [9], [10], [11], [12], [13], [14], [15]. In the laser-MAG hybrid process, metal transfer plays a significant role in determining the arc stability and weld quality [9]. There are three major distinct modes of metal transfer: short-circuiting, globular and spray. When welding current is small and/or arc length is short, the droplet may not be detached until it contacts the pool, resulting in short-circuiting transfer with some explosion. In slightly higher current hybrid welding, the droplet can be stably transferred to the molten pool, and the utilization ratio of laser energy is improved. The droplet transfer mode is affected by the type of the voltages and current intensity range, electrode polarity, distance between the laser-to-arc D_{LA} [10], [11], laser power [10], and shielding gas [14], [15]. Usually, spatters are generated on the surface of the weldment if the above parameter selection is inappropriate.

The frequency of droplet transfer in laser hybrid welding was investigated. The results showed that the thermal radiation of a large number of metal plasma produced by laser, which is promoted the droplet transfer. At the same time, and the droplet transition mode and transition frequency are changed, which is due to the attraction of laser plasma to the droplet and the recoil of the metal vapor to the droplet. The synergetic effect of attraction of laser plasma to the droplet and the recoil of the metal vapor to the droplet effect on droplet transition mode and transition frequency were summarized in these literatures [13], [16], [17]. The investigator provided the above explanation according to the experimental phenomena in laser-MAG hybrid welding. Although researchers have extensively conducted research work on the metal transfer behavior during laser-MAG hybrid welding, the understanding of mode and frequency of droplet transfer is still very limited due to the lack of thorough investigation on how the force state and motion characteristics of metal droplet in the arc space from the perspective of process physics during laser-MAG hybrid welding. However, few researchers have studied the behavior of droplet stable transition in arc space with under laser action.

In order to further investigate the physical mechanism of metal droplet transfer, the high speed camera was used to obtain the in-situ and accurate information of the arc shape and metal droplet of hybrid welding in this work. Based on the information provided by the captured images, the relation between droplet transfer mode and critical current in laser-MAG hybrid welding process was provided. The force and motion characteristics of the droplet in the arc space were given by measuring and mathematical calculation. The calculation results were then used to reveal the behavior of metal droplet transfer during laser-MAG hybrid welding process.

2. Experimental method

2.1. Materials

High strength steel plates with the dimension of 150 mm × 30 mm × 7 mm were used in the study. The specimen surfaces were chemically cleaned by acetone before welding to eliminate surface contamination. Austenitic stainless steel (ASS) filler wire with 1.2 mm in diameter was used. The chemical compositions of the base metal and the filler wires determined by ARL 4460 Metals Analyzer are shown in Table 1.

Table 1. Experimental parameters.

Welding parameters	
Welding speed v [m/min]	0.8
Angle of welding torch θ [°]	60
Defocusing amount Δf [mm]	−2
Extension length L [mm]	12
Wire diameter d [mm]	1.2
Laser powder P [kW]	3.0
MAG current I [A]	150–260
MAG voltage U [V]	24–28
Distance between laser and arc D_{LA} [mm]	0–6

2.2. Welding equipments and conditions

The experimental setup for the pulsed laser-MAG hybrid welding process is shown in Fig. 1, which mainly consists of a 4.0 kW Nd:YAG solid state laser (model HL4006D by TRUMPF) and a metal active gas (MAG) welding system (model YD-350A G2HGE by Panasonic). The Nd:YAG laser with a wavelength of 1.06 µm was operated. During the experiment, the laser beam was focused using a fixed focus lens with a focal length of 220 mm. By placing the sample-material top surface 2 mm above the focal point (Δf = −2 mm) the spot diameter of 0.46 mm was achieved. The spot diameter was measured by the beam quality analyzer (model Focus Monitor 120 by PRIMES). The shielding gas for the arc torch composed of Carbon Dioxide (10 vol%) and Argon (90 vol%) was delivered at a volume flow rate of 16 L/min. The welding experimental parameters are given in Table 1.

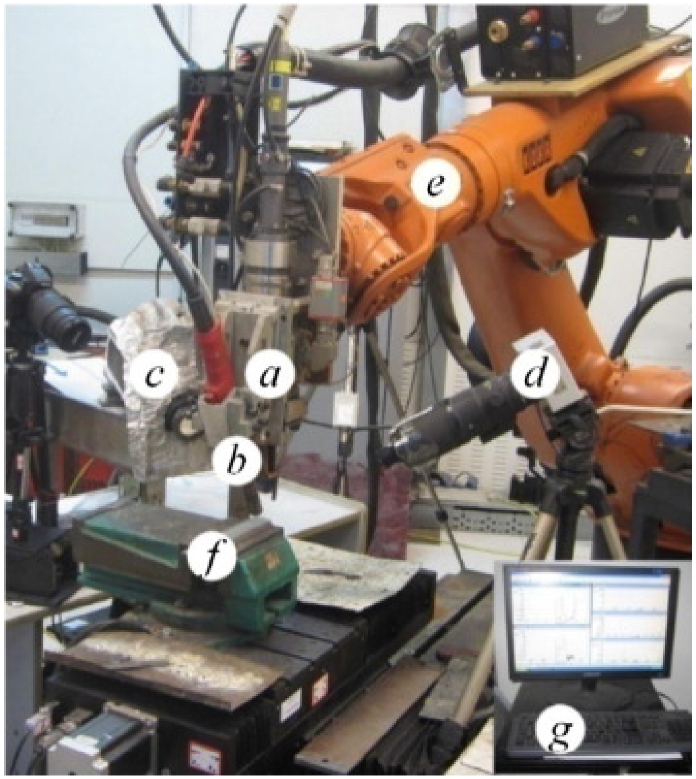

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Fig. 1. (a) Experimental setup: (a) laser beam; (b) MAG torch; (c) high speed camera-1; (d) high speed camera-2; (e) robot; (f) workpiece; (g) welding process analyzing and monitoring system.

The welding arc shape and metal transport phenomena were recorded during the experiments using a CMOS high speed camera (model CR5000×2 from Optronis) operating with a framerate of 4000 fps at full resolution. To collect the shape of arc, a LD pumped green laser was used as backlight source. The LD pumped laser with an emission wavelength of 532 nm can deliver in continuous wave (CW) mode. The interference filter with wavelength of 532 nm was installed in front of high speed camera for filter out most of the arc light. The high speed camera-1 was placed on the side of the hybrid welding head and the angle between the high speed camera and the Y direction was 150° in YZ plane, and equipped with a fast Gigabit ethernet connection to transfer the previously captured high-speed image sequences to a computer. The high speed camera-2 was used to capture droplets.

2.3. Measurements characteristics of arcs and droplet transfer

A high-speed color camera was used to capture shadow images of droplets. In each experimental run, the video image was captured. As schematically shown in Fig. 2, the motion characteristics of the droplet of hybrid welding, defined by its displacement (S_R), velocity (V_R) and acceleration (a), was suitably measured by properly computerized scaling technique, which applied on a number of photographs of droplet falling process. In order to maintain uniformity in comparable measurement, the S_R was measured with the help of the PHOTOSHOP SOFTWARE. The detachment of droplet from the wire end was continuous, which was taken by high speed camera in consecutive frames of video-graphs. The moving distance of the droplet was determined by measuring the distance between the tip of filler wire and the droplet which was shown in Fig. 2. The curves of velocity and acceleration of the droplet were obtained by derivation of the time displacement curves.

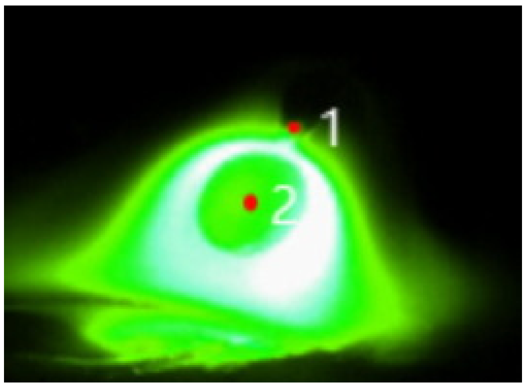

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Fig. 2. Sketch map of displacement measurement of droplet in arc space.

3. Results and discussion

3.1. Mode of droplet transfer and critical current

Fig. 3 shows four transfer modes of short circuiting transfer, globular transfer, projected transfer and spray transfer in laser-MAG hybrid welding. Under the condition of small current and short arc length, the metal transition mode is short circuit transition, as shown in Fig. 4. The nature of an ideal short-circuiting transfer should be essentially an energy determined integrated process consisted of the liquid metal bridge establishing, extension and subsequent instability rupture after metal droplet contacting with molten pool. In the short circuit contact, a powerful electric current is passed through the liquid bridge. Under the influence of the electromagnetic contraction force produced by the powerful electric current, the liquid bridge is broken. At the same time, the liquid bead at the end of the welding wire rises up to form a spatter. This phenomenon can be observed in high-speed photos.

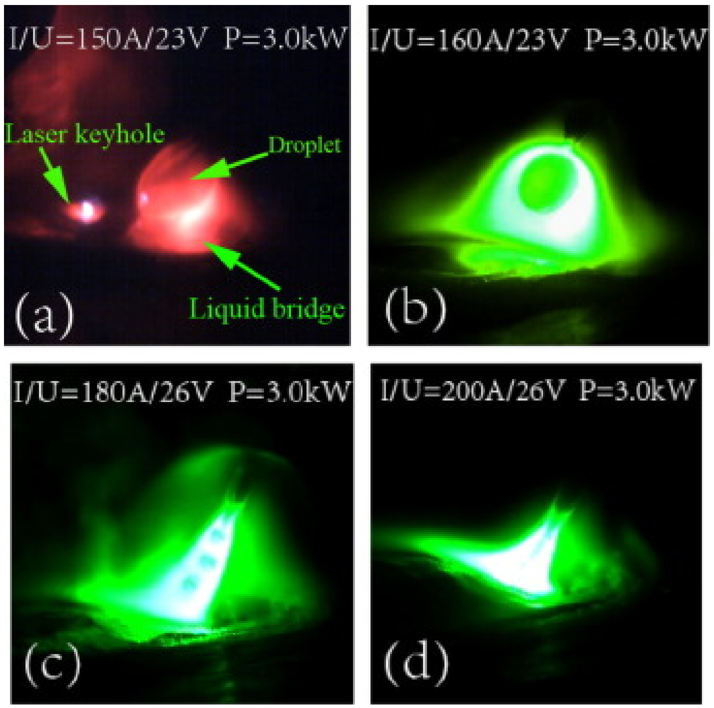

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Fig. 3. Droplet transfer model in laser arc hybrid welding (a) short circuiting transfer (b) globular transfer (c) projected transfer (d) spray transfer.

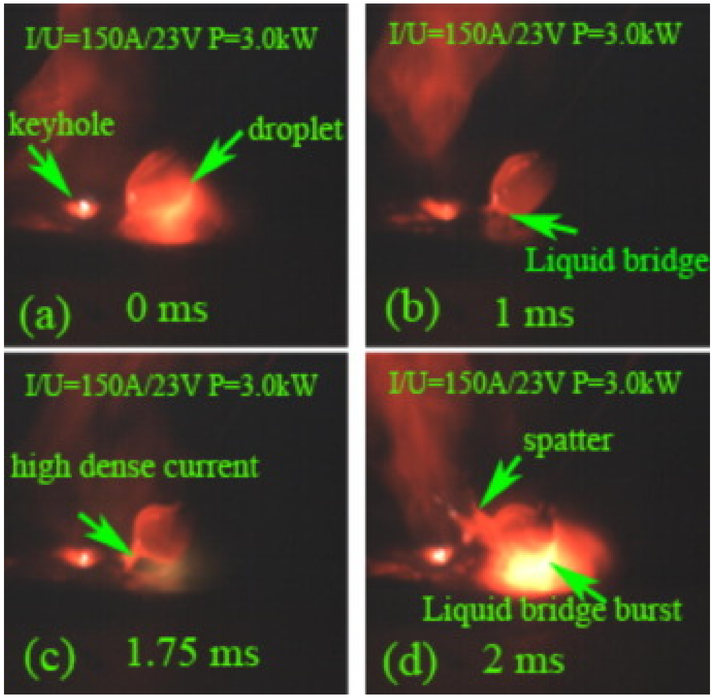

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Fig. 4. Short circuiting transfer in laser arc hybrid welding under low welding current and low voltage.

When the current is near the critical current or greater than the critical current value with a matching low voltage, the arc will produce an inherent self-adjusting characteristic, which will increase the melting speed of the welding wire. When the molten metal contacts with the molten pool, the high current flows through the necking between the droplet and the wire and a consequent neck breaks and forms spatters, as shown in Fig. 5.

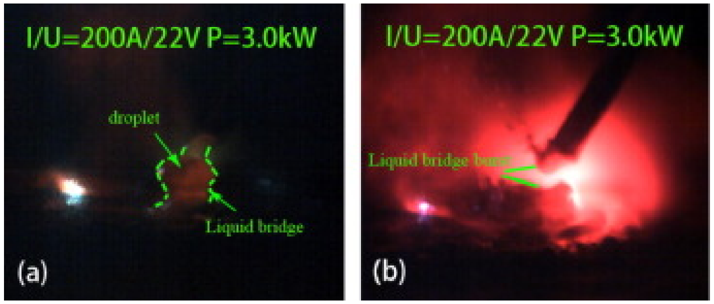

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Fig. 5. Short circuiting transfer in laser arc hybrid welding under high welding current and high voltage.

Fig. 3(b) shows the morphological characteristics of the droplet with transition from the end of the weld wire to the molten pool. When the welding current is small and the arc is long, the molten metal is suspended at the wire end. As the amount of molten metal increases, the droplet will overcome the surface tension and drop to the molten pool. Under the action of the impact force of the droplet falling and the arc force, the weld pool forms a dimple. When the welding current increases, the molten metal at the wire end will show small particles, at the same time, the electromagnetic force produced by the current will be greater than the surface tension of the molten wire, so the particles will be detachment from the ends of the welding wire and hit the weld pool with high speed, as shown in Fig. 3(c). Fig. 3(d) shows the transition image of the droplet at high welding current. The molten metal at the wire end is set up by the electromagnetic contraction force beam and then transferred with high speed to the molten pool. At this time, the wire end like a sharpened pencil.

Fig. 6 shows the relationship between the welding current and the droplet transfer frequency and the droplet diameter. When the welding current is lower than the critical current 150 A, the detachment force of the droplet is less than the detachment resistance. At this point, droplet granulation cannot occur, so that the molten metal at the wire end will contact the weld pool to form a short circuit as shown in Fig. 4. Fig. 6 shows the droplet transfer frequency increases slowly as the welding current increases until 200 A. However, when the welding current is greater than 200 A, the frequency of droplet transfer is increase dramatically. When the welding current increasing from 206 A to 226 A, the droplet transfer frequency increase from 100 drops per second to 700 drops per second, but the droplet diameter decreases from 1.26 mm to 0.75 mm.

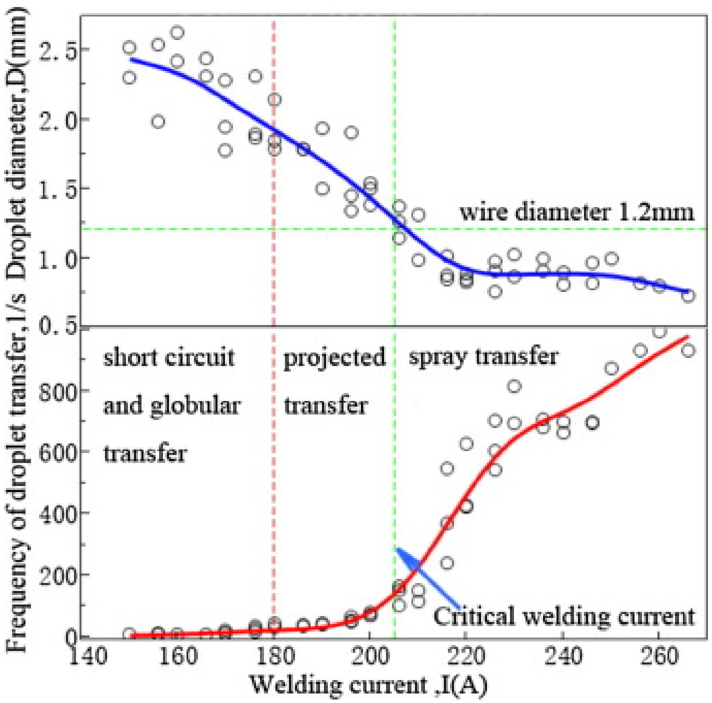

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Fig. 6. Relationship between welding current and droplet transition frequency and droplet diameter.

3.2. Motion characteristics of droplets in arc space

Fig. 7 shows droplets with similar spatial dimensions of arc welding and laser arc hybrid welding respectively. Fig. 8 shows the time displacement, velocity and acceleration curve of droplet in the arc space, which is detachment from the wire end for different welding processes. It can be seen from Fig. 8(a) that the observed displacement contains some vibration component, which is caused by the vibration motion of the droplet when it is trying to recover the sphere in the arc space. Fig. 8(b) shows the instantaneous velocity of the droplet transfer obtained in accordance with Fig. 8(a). It can be seen from Fig. 8(b) that the instantaneous velocity of droplet decreases first and then increases with the increase of time for MAG welding, but the instantaneous velocity of droplet increases with the increase of time for laser-arc hybrid welding. Fig. 8(c) shows the acceleration of the droplet which is detachment from the wire. It can be seen from the figure that the acceleration of the droplet in MAG welding increases gradually, while the acceleration of droplet in laser arc hybrid welding is very small. This indicates that the effect of complex force on droplet in the arc space for laser arc hybrid welding. The acceleration of the droplet detachment from wire end is 70 m/s^2 and 50 m/s^2 for arc welding and laser arc hybrid welding separately. These values are 7 times and 5 times than the acceleration of gravity.

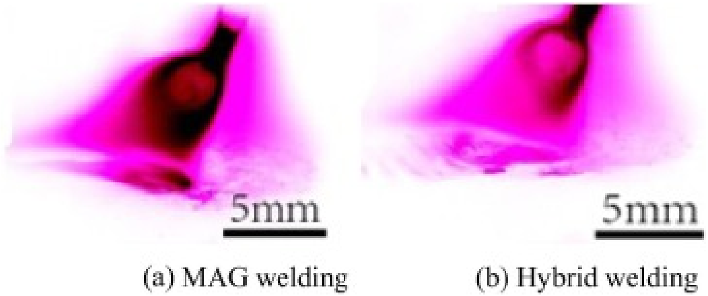

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Fig. 7. Droplet diameter.

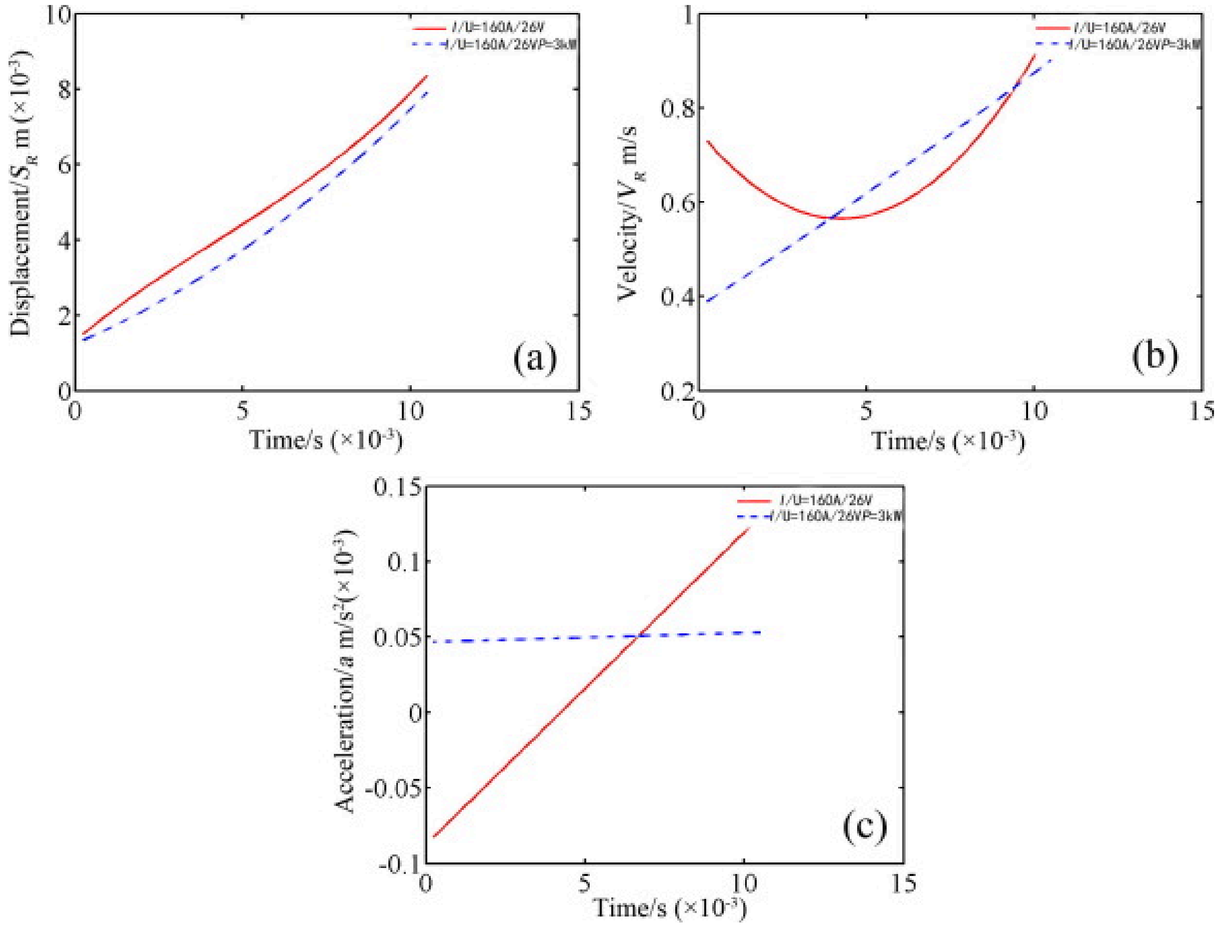

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Fig. 8. Motion characteristics of droplets in arc space: (a) displacement (b) velocity (c) acceleration.

According to the above observed acceleration, the detachment force of the droplet F_A and F_{Hy} respectively:

$$F_A = ma = \frac{4}{3}\pi r^3 \rho_m a = \frac{4}{3}\pi \times 0.115^3\ \mathrm{cm}^3 \times 7.1\ \mathrm{g/cm}^3 \times 7000\ \mathrm{cm/s}^2 = 316\ \mathrm{N} \tag{1}$$

$$F_{Hy} = ma = \frac{4}{3}\pi r^3 \rho_m a = \frac{4}{3}\pi \times 0.115^3\ \mathrm{cm}^3 \times 7.1\ \mathrm{g/cm}^3 \times 5000\ \mathrm{cm/s}^2 = 226\ \mathrm{N} \tag{2}$$

From the calculation results, it can be seen that the resultant force is smaller along the direction of acceleration for laser arc hybrid welding when the droplet moves in the arc space. The results show that the stress state of the droplet is changed by the introduction of the laser.

3.3. Force analysis of droplet in arc space

From the picture of droplet transfer, it can be seen that the droplets are formed, fallen and entered into the molten pool, which is accompanied by a combination of forces. In this paper, the forces acting on the droplets in the arc space are discussed. According to the above results, the acceleration of the droplet in the arc space during the laser arc hybrid welding process can reach 5 times of the acceleration of gravity, which is less than the acceleration of the droplet during the arc welding. Fig. 9 shows a schematic diagram of the external force after the droplet is detachment from the wire end. The forces acting on the droplets in the arc space are mainly gravity, plasma flow force, electromagnetic contraction force, and metal vapor reaction force.

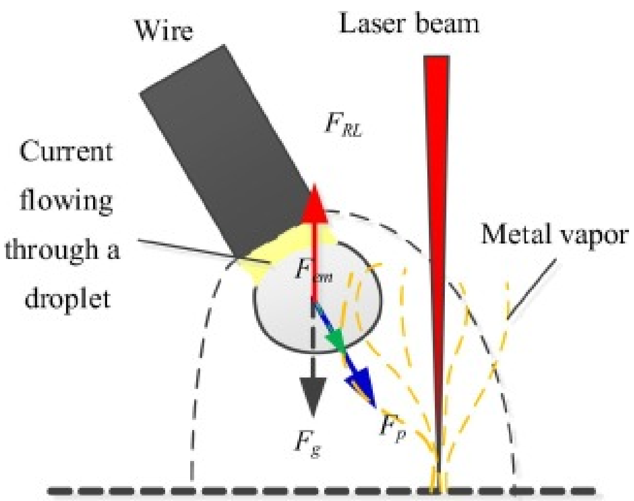

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Fig. 9. Schematic diagram of the force acting on the droplet just off the end of the wire.

In the arc space, the gravity acting on the droplets is represented by the following equations [18]:

$$F_g = \frac{4}{3}\pi R^3 \rho g \tag{3}$$

where R is the droplet radius, ρ is the droplet density, and g is the gravity acceleration. It can be seen from the formula that the gravity of the droplet is proportional to the diameter of the droplet.

The plasma flow force (F_p) plays an important role in process of the droplet transfer. The high temperature air flow (plasma flow) in the arc space flows rapidly toward the molten pool from the wire end. The droplet is affected by the plasma flow, which has an important influence on the detachment of droplets and the state of motion after detachment. The F_p can be expressed by Eq. (4) [18].

$$F_p = \frac{C_d A_p \rho_f v_f^2}{2} \tag{4}$$

where C_d is the plasma flow coefficient (C_d=0.45), A_p is the action area of the plasma flow, v_f is the velocity of the plasma flow, ρ_f is the density of the plasma flow.

The electromagnetic contraction force (F_{em}) is an important force which is affecting the droplet transfer. The electromagnetic contractile force is the macroscopic reflection of the Lorenz force in the magnetic field by the current in the droplet. When the current in the droplet is dispersed, the Lorenz force acts as a detachment force, however, during the current in the droplet is converged, the Lorenz force prevented the resistance force.

The electromagnetic contraction force can also be expressed by the parameters of the current in the droplet, as shown in Eq. (5) [12].

$$F_{em} = \frac{\mu_0 I^2}{4\pi} F_{emz} \tag{5}$$

$$F_{emz} = -\left[\frac{1}{4} - \ln\left(\frac{r_d \sin\theta}{r_w}\right) + \frac{1}{1-\cos\theta} - \frac{2}{(1-\cos\theta)^2}\ln\left(\frac{2}{1+\cos\theta}\right)\right] \tag{6}$$

where I is welding current, r_d droplet radius, r_w wire radius, θ arc hanging angle and μ_0 is the permeability of free space ($4\pi \times 10^{-7}$ Hm^{-1}). The properties and sizes of the electromagnetic force of different welding current can be referred to literature [13].

In the laser welding process, when the laser beam irradiated vertically on the material surface, the material absorbs laser energy led to solid metal rapid heating, melting, vaporizing, forming laser keyhole and spurting out a lot of metal vapor form laser keyhole. The physical model of the removing molten metals is shown in Fig. 10. When laser-MAG hybrid welding, the metal vapor ejected from the laser keyhole produces a reaction force to the droplet.

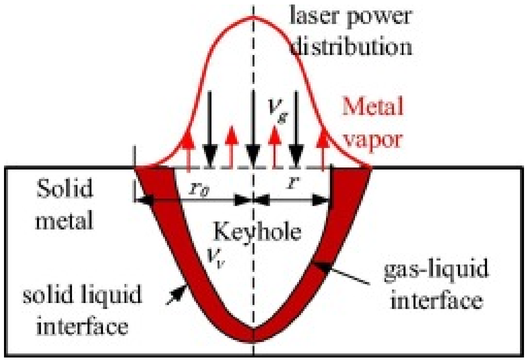

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Fig. 10. A physical model for removing molten metals from an interaction region.

For the Nd:YAG laser, it is assumed that the energy distribution is uniform in a circular spot with a radius of R. The effective incident radius of laser beam energy is the focal spot radius of laser beam. Assume that the metal vapor from the laser keyhole is incompressible fluid. According to the theory of gas dynamics, the evaporation of molten metal and the flow of metal vapor should satisfy the conditions of conservation of mass flow [19]:

$$\rho_m v_v = \rho_g v_g \tag{7}$$

r - Beam radius; ρ_m – Liquid metal density; v_v - The rate at which the liquid zone evaporates the front wall; ρ_g - Metal vapor density; v_g - Rate of metal vapor ejection.

The metal vapor reaction force on the droplet is equivalent to the flow resistance of the object immersed in the moving fluid, which is caused by the relative motion between the object and the fluid. In a uniform velocity fluid, the reaction force of metal vapor on the droplet can be expressed according to Stokes's theory:

$$F_D = C_D A \frac{\rho_g v_g^2}{2} \tag{8}$$

where C_D is the coefficient of flow resistance related to the Reynold number of metal vapor (C_D=0.45); A is the projected area of the object perpendicular to the direction of flow.

It is assumed that metal vapor can be regarded as an ideal gas. According to the Clapeyron equation of the ideal gas, the pressure of saturated metal vapor can be expressed by Eq. (9):

$$P_s = \rho_g R_0 T_S = \frac{N_a k_B}{M_a} \rho_g T_S \tag{9}$$

$$\rho_g = \left(\frac{M_a}{N_a k_B} \right) \frac{P_S}{T_S} \tag{10}$$

where R_0 is the ideal gas constant, relating to molecular weight M_a of gas; $\mathrm{N_a}$ is Avogadro constant (6.02×10^{23}/mol); $\mathrm{k_B}$ is the Boltzmann constant ($1.38 \times 10^{-23}\,\mathrm{m^2\,kg\,s^{-2}\,k^{-1}}$); $\mathrm{T_S}$ is the surface temperature of the molten metal.

In document [20], the Semark Model was introduced. The rate of evaporation of the front wall of liquid region is determined by the surface temperature of the molten metal:

$$v_v = V_0 \exp(-U/T_S) \tag{11}$$

where $\mathrm{V_0}$ is a constant, the general value is 3.4×10^2 m/s.

When the incident laser energy is stronger, the laser energy absorbed on the material surface leads to the increase of the temperature of the molten body, and the stronger the reaction force of the metal vapor on droplet. The pressure value (PS) of the saturated vapor pressure depends on the surface temperature of the molten body. As the temperature of the liquid increases, the activity and evaporation capacity of the molecules increase, so the saturated vapor pressure of the liquid increases. According to the Semark model, the pressure of the saturated vapor pressure can be expressed as Eq. (12):

$$P_S = P_S(T_S) = B_0 T_S^{-1/2} \exp(-U/T_S) \tag{12}$$

where U is the calculated value, $U = M_a L_v / (N_a k_B)$; L_v is the latent heat of vaporization of a molten body; $\mathrm{B_0}$ is the evaporation constant of the molten body.

The v_g can be obtained by simultaneous Eqs. (7), (10), (11):

$$v_g = \rho_m v_v \left(\frac{N_a k_B}{M_a}\right)\frac{T_S}{P_S} = \rho_m V_0 \left(\frac{N_a k_B}{M_a B_0}\right) T_S^{\frac{3}{2}} \tag{13}$$

Eqs. (10), (13) are substituted into (8), and Eq. (14) can be obtained. At this point, assuming that the metal vapor produced in the laser keyhole is a uniform velocity flow, Eq. (12) can indicate the reaction force of the metal vapor on the droplet.

$$F_D = \frac{1}{2} C_D A \rho_m^2 V_0^2 \frac{N_a k_B T_S^{3/2}}{M_a B_0} \exp(-U/T_S) \tag{14}$$

The characteristics of droplet transfer in laser-MAG hybrid welding process were investigated in the literature [9], [11], [12], [13], and it was considered that the metal vapor had a hindrance effect on the droplet transfer. The distance between the laser keyhole and the position of the droplet falling into the molten pool is 0, the laser beam acts on the droplet and causes the welding process to be unstable. In the actual laser arc hybrid heat source welding process, there is a certain distance (D_{LA}) between the laser keyhole and the position of the droplet falling into the molten pool. Therefore, the reaction force of metal vapor on droplet and its range of action are assumed as follows: The action mode of laser heat source is Gauss rotating body heat source [21].

$$\mathrm{q}(r,z) = q_m(0,z) \cdot \exp\left[-\frac{3r^2}{r_0^2(z)}\right] \tag{15}$$

The area radius of heat flow is $r_0(z)$. Along the thickness direction of weldment, the radius $r_0(z)$ of heat flow zone shows regular attenuation. However, the distribution of the metal vapor flow generated by laser welding is represented by Eq. (16):

$$F_D = \frac{1}{2} C_D A \rho_m^2 V_0^2 \frac{N_a k_B T_S^{3/2}}{M_a B_0} \exp(-U/T_S) \cdot \exp\left(-\frac{3r^2}{r_0^2(z)}\right) \tag{16}$$

The relation curve of the metal vapor recoil with the change of heat source distance is calculated by Eq. (16) as shown in Fig. 11. As can be seen from the diagram, the metal vapor recoil force is 60 N for the distance equal to 0 mm; during the distance between the heat sources is 3 mm, the metal vapor recoil force is roughly equivalent to 0. It can be seen that in actual laser arc hybrid welding, the metal vapor recoil force acting on the droplet is very small. That is to say, the metal vapor ejected by the laser keyhole has little influence on the droplet transfer. The transition mode, transition frequency and stress state of droplet in laser arc hybrid welding are expounded in Refs. [12], [13]. It indicates that the introduction of laser reduces the frequency of droplet transfer. In this paper, the mathematical calculations are given to show that the metal vapor produced by the laser has little influence on the mode and frequency of droplet transfer. So, how does the laser produced plasma affect the mode of droplet transfer and the frequency of droplet transfer? The calculation results show that the acceleration of droplet in hybrid plasma arc is almost the same as shown in Fig. 7. This indicates that the resultant force of the droplets in the hybrid plasma arc is less than that in the arc plasma arc. In other words, the electromagnetic resistance acting on the droplets increases. From Fig. 12, it can be seen than the introduction of the laser obviously compresses arc, especially the arc near the weld pool produces more intense contraction. This result directly causes change of the electromagnetic contraction force and the plasma flow force acting on the droplet. The resultant force of droplet decreases with increasing electromagnetic resistance in hybrid arc space. In other words, the electromagnetic resistance increases in the hybrid arc space, which is due to the arc contraction. Fig. 13 shows the transition image of adjacent droplets. The droplet transfer is blocked near the molten pool and is combined with the subsequent drops droplet as shown in Fig. 13(b). Finally, the laser plasma compresses the arc and leads to the increase of the electromagnetic resistance of the droplet transfer, which leads to the change of the droplet shape and the decrease of the transition frequency.

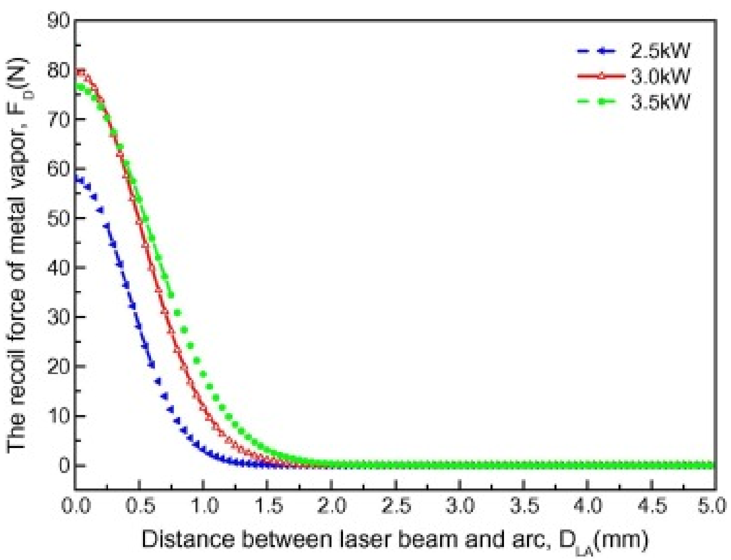

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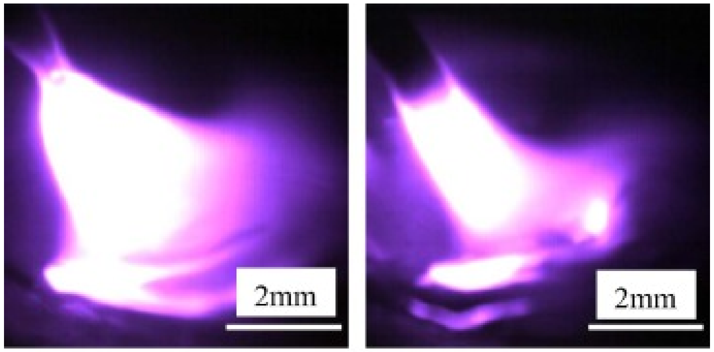
Fig. 11. The relation between (D_{LA}) and the reaction of metal vapor (F_D).



(a) Arc welding (b) hybrid welding

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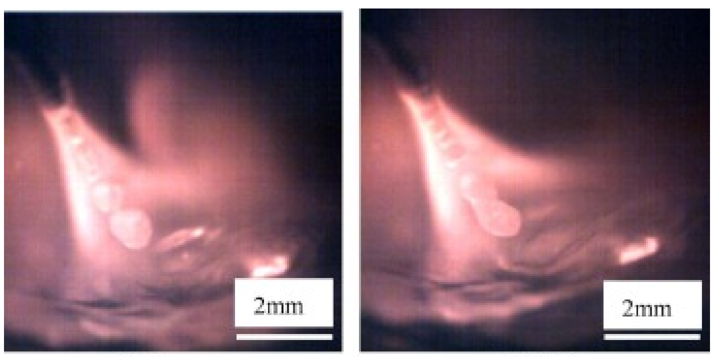
Fig. 12. Plasma morphology under different welding methods.



(a) 0ms (b) 0.2ms

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Fig. 13. Droplet coalescence during droplet transfer.

4. Conclusions

The droplet transfer behaviors and the droplet stress state for MAG welding and laser-MAG hybrid welding were investigated in this study. The critical welding current of globular transition mode is 180 A. The droplet transfer mode changes from projected transfer to spray transfer with the increasing of the welding current. In laser-MAG hybrid welding, the laser plasma compresses the arc and leads to the increase of the electromagnetic resistance of the droplet transfer, which leads to the change of the droplet shape and the decrease of the transition frequency. The metal vapor recoil force acting on the droplet is calculated by the mathematical calculations. When the distance between the droplet and the laser keyhole is 3 mm, the acting force of metal vapor from the laser keyhole on the droplet is very small.

Conflict of interest

The authors do not have any possible conflicts of interest.

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Appendix A. Supplementary material

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Supplementary data 1.

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